PH227: AI and Data Science

Tutorial Sheet 4: Python/Data-Science Part

Note: This tutorial sheet contains programming problems related to various concepts in statistics discussed in the lectures. Each tutorial sheet is worth 20 marks, and you have to submit the codes corresponding to the starred (*) problems to the TAs. The marks of the starred problems are indicated next to them.

1. * Write a Python program to generate N random numbers distributed according to the normal distribution with chosen mean (μ) and standard deviation (σ), and to make a histogram plot of those numbers using the hist() function of matplotlib. The random number generation was practiced in detail in the jupyter notebook of Lesson-3, and note that N, μ, and σ are to be read from the input. Verify that as N increases, the histogram plot begins to resemble the normal distribution more and more. (5 marks)

2. Consider the random numbers of the previous problem. Write a Python program, using for/while loops, to calculate the mean, median, standard deviation, skewness and kurtosis of these numbers. Compute and print out the same quantities using appropriate numpy (for mean, median, and standard deviation) and scipy (skewness and kurtosis) functions to verify that your program is correct.

3. * Consider the Auto.csv file discussed in jupyter notebook Lesson-7. After removing the rows corresponding to the missing values (listed as ?), calculate the mean, median, standard deviation, skewness and kurtosis of the numbers listed in the following columns, using your own code, and the corresponding numpy and scipy functions

   (a) # 3, which has the engine displacement data

   (b) # 4, which has the horsepower data.

   (c) Additionally, make the histogram plots for the data in both the columns.

4. * The problem is about a numerical verification of the "Central Limit Theorem" (CLT).

   Program Task: Write a program that performs the following steps:

   - Generate a large population of at least 100,000 random numbers from a uniform distribution on the interval $[0, 100]$.
   - Calculate the theoretical population mean μ and standard deviation σ. For a uniform distribution on $[a, b]$, $\mu = \frac{a+b}{2}$ and $\sigma = \sqrt{\frac{(b-a)^2}{12}}$.
   - Create an empty list to store sample means.
   - Repeat the following process 10,000 times:
     - Draw a random sample of size $n = 30$ from your generated population.
     - Calculate the mean of this sample.
     - Add the calculated sample mean to your list mentioned above.

   Analysis and Visualization:

   - Create a histogram of your original population data. Describe its shape.
   - Create a histogram of the list of 10,000 sample means. Describe the shape of this new distribution. How does it compare to the original distribution?

- Calculate the mean and standard deviation of your list of sample means (the empirical values).
- Compare these empirical values to the theoretical predictions from the Central Limit Theorem ($\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$). (10 Marks)

Note: Note that in numpy there are functions using which one can draw m random samples from a bigger list containing $N \gg m$ entries. For our problem, $m = 30$, and $N = 100000$.